건축과경관디자인 중간고사 · 출력용 정리

10/14(수) 15:10–16:00 · 54-120 · 영어 (매칭 / T·F / 단답 / 서술)

① 타임테이블 (밤샘 버전 · 학교)

시간	할 일
☐ 01:30–02:15	서술형 3개 암기 — 소리 내어 2회 + 키워드 보고 말로 재생 (p.2)
☐ 02:15–03:00	3주차 전체 + Lynch 5요소 (p.3–4) 🔴
03:00–03:10	휴식 (걷기·물)
☐ 03:10–03:50	4주차: 7원칙, 모더니즘, Square vs Plaza, 오픈스페이스 (p.5) 🔴
☐ 03:50–04:25	7주차: Active 4단계, 출입구, 공개 공지 6유형, 수치 (p.6) 🟡
☐ 04:25–04:45	1·2·5주차 + 수치·인물 표 (p.3, p.6)
☐ 04:45–05:15	Claude에게 "2차" → 퀴즈 → 오답 복습
☐ 05:15–05:45	서술형 3개 영어로 실제로 쓰기 (각 8분, p.2 키워드만 보고)
05:45–06:00	p.1 함정표 + p.3 수치표 훑고 바로 눕기
06:00–07:30	쪽잠 90분 (수면 1주기 · 알람 2개) — 자기 직전 본 내용이 잘 기억됨
07:30–08:00	세수·아침·커피
☐ 08:00–09:00	기억 재생: Lynch·인물·수치를 안 보고 백지에 써보기 → 틀린 것만 확인
09:00–11:30	다른 수업 (쉬는 시간: p.3)
11:30–12:00	점심
12:00–12:20	파워냅 20분 (25분 넘기지 말 것)
☐ 12:20–13:20	1차·2차 오답 위주로 PDF 재확인
☐ 13:20–13:50	Claude에게 "3차" → 실전형 최종 모의고사
☐ 13:50–14:30	오답 복습 + 서술형 마지막 1회 쓰기
14:30–15:00	p.1–3 훑기 → 이동 (카페인은 14:00 이전까지만)

② 1차 모의고사에서 틀린 함정 (T/F 주의!)

내 경향: 헷갈리면 T를 고름 → 틀린 4개가 전부 T였음. 문장 속 핵심 명사 하나가 바뀌었는지 확인하고, 근거가 없으면 T로 찍지 말 것.

함정 문장 (F)	정답
Mapping = drawing lines of movement	그건 Tracing. Mapping = 머무는 위치(stays)를 기호 X, O, ▲로 표시
Plaza = formal space at focal point of civic importance	그건 Square. Plaza = 상업(오피스·리테일) 건물에 딸린 공간
Private residential entrance = flush with sidewalk	Flush는 공공 출입구. 주거 = 높이 차 ≤0.9m, 이격 ≥3m
Mirrored/frosted glass enables communication	투명 유리가 소통. 미러·반투명은 안→밖 조망만
Min. ground floor height 3.0m	4.5m (하역 공간 높이 확보)
Necessary activities only in good weather	모든 조건에서 발생. 좋은 환경이 필요한 건 Optional
People prefer stairs to ramps	Ramps 선호
Affordance = 편안함을 느끼는 심리적 영역	그건 Personal space(Hall). Affordance = 행동 가능성(Gibson)

T였던 것: Le Corbusier 오픈스페이스 85–95% · Lynch Identity/Structure/Meaning · 5–6m 폭·100m당 15–20개 점포 · Stepped approach(임시→영구) · Context-less design → "anywhere places"

③ 서술형 모범답안 (외워서 쓰기)

E1. Affordance

Affordance, proposed by James J. Gibson (1977), is the possibility for action that an object or environment offers to a perceiver. It depends on both the perceiver and the object — a tree offers shade to people but shelter to birds. People often do not follow the designer's intention; they find their own affordances, such as leaning on a column or sitting on a handrail or ledge. Therefore design controls usage: e.g. (1) adding a simple curve to a billboard turns it into a bench or shelter (IBM "Smart Ideas"), (2) colored paving indicates walking direction, and painting stair risers motivates people to climb stairs. Good design anticipates these affordances, such as providing seating along edges (edge effect) and following people's direct lines of walking.

E2. Active vs. Inactive Ground Floor

According to Jan Gehl, an active ground floor has small units with many doors, large variation in function, no blind units, rich façade relief and good details. An inactive one has large units with few or no doors, no visible variation in function, blind or passive units and uniform façades with nothing to look at. (Between them: Mixture, Boring.) Strategies for active frontage: (1) narrow units of 5–6 m, giving 15–20 shops per 100 m — a new experience every 4–5 seconds; (2) transparent glass and maximized windows and doors instead of mirrored or frosted glass, which also adds natural surveillance; (3) a generous first-floor height of at least 4.5 m; (4) canopies, awnings, balconies and "spill-out" space such as sidewalk cafés and large folding doors that connect inside and outside.

E3. Criticism of Modernism

Le Corbusier's Contemporary City for Three Million Inhabitants (1922) proposed a rigid geometric, centralized plan, monumental axes, uniform modern buildings ("towers in the park"), 85–95% open space and a highway-based traffic system (also Chandigarh). Modernists rejected the city and city space and focused on individual buildings, so human scale was forgotten. The result was large, cold spaces with few people, and car-invaded cities with noise, pollution and accident risk. Jan Gehl criticizes that Modernism put a low priority on public space, pedestrianism and the city's role as a meeting place. He argues that architects and planners must reinforce pedestrianism to create lively, safe, sustainable and healthy cities — "First we shape the cities, then they shape us."

14:35 쓰기 연습용 키워드만

E1	Gibson 1977 · action possibility · perceiver + object · tree: shade/shelter · lean on column, sit on handrail · design controls usage · IBM curved billboard · colored paving · stair risers · edge effect · direct lines
E2	Gehl · small units/many doors vs large/few · variation · blind · relief · 5–6m, 15–20/100m, 4–5 sec · transparent glass, surveillance · 4.5m · canopy/awning/balcony · spill-out, café, folding doors
E3	Le Corbusier 1922 · 3 million · geometric, axes, uniform · towers in the park · 85–95% · highways · rejected city space → buildings · human scale forgotten · cold spaces · cars · Gehl: low priority on public space · lively/safe/sustainable/healthy

④ 수치 표

수치	의미
0–0.5 / 0.5–1.2 / 1.2–3.6 / >3.6 m	Hall: Intimate / Personal / Social / Public distance
100 m / 25 m / 7.5 m	Social field of vision: 움직이는 사람이 보임 / 표정·감정 읽음 / 모든 감각 작동
85–95 %	Le Corbusier 1922 계획의 오픈스페이스 비율
5–6 m · 15–20개/100 m · 4–5초	활기찬 상업가로 점포 입면 폭 · 점포 수 · 새로운 경험 간격
4.5 m	1층 층고 최소 (floor-to-floor)
≤0.9 m / ≥3 m	주거 출입구 높이 차 / 대지경계선에서 이격
10초	보행자 횡단 대기 기준 (Convenient)
2.5×5 m / 2×6 m	45° 사선 주차 / 평행 주차

⑤ 인물 – 이론

인물	키워드
Edward T. Hall	Proxemics, Personal space, 4 distances · *The Hidden Dimension* (1966)
James J. Gibson	Affordance (1977) "Afford + -ance"
Kevin Lynch	*The Image of the City* (1960) · Legibility · 5 elements · Boston
Jan Gehl	*Cities for People* (2010) · 3 activities · Edge/Piano effect · Active frontage · modernism 비판
Gehl & Birgitte Svarre	*How to Study Public Life* · 5 questions · survey methods
Jane Jacobs	*The Death and Life of Great American Cities* · "If a city's streets look interesting, the city looks interesting"
Jacobs & Appleyard (1987)	*Toward an Urban Design Manifesto* · prerequisites for a sound urban environment
Richard Rogers	"Cities are the places where people meet to exchange ideas, trade, or simply relax"
Le Corbusier	Contemporary City for 3 million (1922) · Chandigarh · towers in the park
Robert Sommer	*Personal Space: The Behavioral Basis of Design* (1969)
Yoshinobu Ashihara	*The Aesthetic Townscape* (1983)
Llewelyn-Davies	*Urban Design Compendium* (2000) — 7원칙 출처
Isoya / Matsumoto	좌석 점유 실험 (지하철 끝자리부터, 소변기는 입구에서 먼 것 선호)
Meejin Yoon	Mobius dress (personal space 가변)

⑥ Lynch 5요소 (1차에서 1/5 → 꼭!)

요소	한 줄 공식	정의	예시
Paths	움직이는 선	Channels along which the observer moves · 이미지의 지배적 요소 · 다른 요소를 연결	streets, walkways, transit lines, canals, railroads
Edges	경계 선 (안 다님)	Linear elements not used as paths · boundaries between two phases · barriers or seams	shores, walls, railroad cuts
Districts	들어가는 면	Medium-to-large sections, 2D extent · observer mentally enters "inside of" · common character	historic district, housing complex, Central Park
Nodes	들어가는 점	Strategic spots into which one travels · junctions / concentrations	enclosed square, crossing of paths, street-corner hangout
Landmarks	밖에서 보는 점	Point-references external to the observer · scale varies	tower, high-rise, sign, tree, store front

Environmental image 3요소: Identity(구별) · Structure(관계) · Meaning(의미) | Legibility = apparent clarity of the cityscape → way-finding, security

⑦ 3주차 · How to Study Public Life + Understanding Cityscape 🔴

5 Questions (Gehl & Svarre)

How many?	가장 기본 · counting → quantitative data · 보행량(moving) / 체류(staying) · before & after 비교 · 날씨·시간 기록
Who?	gender, age는 관찰 가능 / 직업·경제상황은 questionnaire 필요
Where?	edges / middle / evenly? public · semi-public · private zone?
What?	활동 유형 mapping: Necessary / Optional / Social
How long?	이동 시간, 체류 시간, 활동 지속 · 좋은 장소일수록 천천히 걷고 오래 머문다

Field Survey Methods (7)

Photographing	"One picture = 1000 words" · time-lapse로 시간 변화
Counting	basic · 모든 것을 셀 수 있음 (인원, 성별, 대화, 미소, 휴대폰, 벤치 수…) · 예외 상황(시위, 공사) 기록
Mapping	머무는 위치(stays)를 평면에 기호(X, O, ▲)로 · 시간대별 레이어 · 한 순간의 그림
Tracing	평면에 이동선을 그림 · 방향 선택, 흐름, 어느 출입구 · GPS
Tracking (Shadowing)	선택한 사람을 따라가며 stop-watch로 측정 · 보행 속도
Keeping a diary	관련 있는 모든 것을 기록 · 공공생활과 공간의 상호작용 디테일
Test walks	관찰자가 직접 주요 경로를 걸으며 대기 시간·장애물·우회 기록

Case study 분석

Existing conditions diagram = 물리적 요소 중심, 축척·north arrow·범례 / Analysis diagram = 이용 패턴·감각적 영향(흐름, 조망, 그늘, 갈등)을 선·화살표로

Lynch – Public Image

Environmental image = observer ↔ environment two-way process. Public image = 다수 시민이 공유하는 common mental pictures.

Boston 문제 분석 예: direction ambiguity, characterless path, elastic intersection, weak boundary, bottomless tower, disconnected hidden waterfront, discontinuity

⑧ 4주차 · Urban Context & Structure 🔴

Modernist vision → 비판

Le Corbusier 1922: rigid geometric centralized orthogonal plan · monumental axes · uniform buildings · 85–95% open space · highways · Towers in the Park · Chandigarh → human scale forgotten, large cold spaces, car-invaded city, low priority on public space & pedestrianism

Urban design = "the process of making better places for people than would otherwise be produced"

7 Principles (Urban Design Compendium) — 원칙 ↔ 키워드

#	Principle	Key aspect	핵심
1	Places for People	Quality of public space, Continuity & Enclosure	safe, comfortable, varied, attractive
2	Enrich the Existing	Character, History, Storytelling	own identity, 모든 스케일
3	Make Connections	Ease of movement, Legibility	permeability, people before traffic
4	Work with the Landscape	Climate, landform, landscape, ecology	natural ↔ man-made 균형
5	Mix Uses and Forms	Diversity	variety & choice (Roppongi Hills, Midtown)
6	Manage the Investment	Sustainable land value	economically viable, maintained
7	Design for Change	Adaptability	flexible (Hammarby Sjöstad, Greenwich Millennium Village)

암기: P-E-C-L-M-I-D "People Enjoy Connected Landscapes, Mixed, Invested, Dynamic"

Scale

Region → City/Town/Village → District → Neighborhood → Block/Street → Building

Process (5 steps)

1. Set Policy (sowing the seed)
2. Undertake Design
3. Attract Investment
4. Secure Planning & Technical Approvals
5. Ensure a Legacy

Context importance

strengthening communities · distinctive places (context-less → "anywhere places") · site assets · integrating · feasibility · vision

Urban structure 중요한 이유 (5)

1. Integration
2. Functional efficiency
3. Environmental harmony
4. A sense of place
5. Commercial viability

Street network

Cul-de-sac = introverted layout, fails to integrate ↔ 기존·신규 가로 연결이 pedestrian-friendly

Open Space Typology

Greenway	footpath·cycle 네트워크, 하천·폐선 따라
Waterfront	lakes, rivers, canals – recreational value
Playing field	active recreation (soccer)
Park	Regional > Metropolitan > District > Local > Small
Square	formal, focal points of civic importance, hard paved, passive recreation
Plaza	associated with commercial (office/retail) buildings, formal landscaping
Playground	child's play, close walking distance to houses

Streets

Community function: Ceremonial/Symbolic · Social space · Place of commerce · Outdoor room | Transportation function: Transit · Vehicles · Bicycles · Walking · Parking

Great streets 요소: Vegetation · Enclosure (many buildings) · Details · Places (wide sidewalks) · Density of people · Diversity of land use · Public transit

⑨ 7주차 · Detailing the Place 🟡

Active	small units, many doors · large variation in function · no blind units · lots of façade relief · good details
Mixture	large & small units · modest variation · some blind units · modest relief · few details
Boring	large units, few doors · almost no variation · many blind units · few or no details
Inactive	large units, few or no doors · no visible variation · blind/passive · uniform façades, nothing to look at

Entrances

- Primary entrance는 public street에 면하고 universally accessible
- 리테일 점포마다 보도에서 바로 출입
- 계단·램프는 Frontage zone 안에
- 공공 출입구 = flush / 주거 = ≤0.9m, ≥3m
- Large folding/retractable doors · 바닥 재질로 입구 강조

Façade

- 창·문 최대화, transparent glass → natural surveillance
- Balconies, canopies, bay windows, awnings
- Spill-out space, sidewalk cafés
- Green walls: energy ↓ , runoff ↓ , air quality ↑

Publicly Accessible Open Space (6)

- Courtyard: 블록 중앙, 가로 접면 없음
- Forecourt: 보도와 주출입구 사이
- Landscaped setback: 보도–건물면 사이
- Plaza: hard surface, 가로에 면한 활기찬 장소
- Urban garden: intimate scale, 정오 일조
- Walkway: 블록 관통 보행로 (유리로 덮으면 galleria)

Street furniture

- Seating: primary(bench, chair) + secondary(stairs, statue bases) · 최적 = 보행흐름 밖, 나무 아래, 사람 구경 가능 · movable chairs
- Bollards · Transit shelter(투명, 점포 사이) · Street lights · Bike racks(고정, 용량 ↑)
- Signage: 구조를 legible하게 → 표지 최소화
- Public art: Cloud Gate(Chicago), Maman(Roppongi)

Floorscape: local materials, historical associations, symbols in pavement | Permeable paving: flooding ↓ , water quality ↑

⑩ 1·2·5주차 빠른 복습 🟢

1주차

Wanted city: Lively · Safe · Sustainable(green mobility) · Healthy
Jacobs & Appleyard prerequisites: livable streets · minimum density · integration of activities · public space defined by buildings · many distinct buildings

2주차

- Gehl 3 activities: Necessary(모든 조건) / Optional(좋은 질 필요) / Social(passive see·hear + active)
- Personal space = body buffer zone, portable territory
- 친밀할수록 가까이 앉음 · 옆에서의 접근에 관대
- Edge effect / Piano effect / Niches
- Direct lines, ramps > stairs
- Paris 높이 규제 · Tokyo 산 조망 · New York 고층 vs 공원

5주차

- 3 realms: Private / Pedestrian / Roadway
- Pedestrian realm: Frontage · Walking(sidewalk) · Landscape(furnishing) · Edge zone
- 5C: Connective · Convenient(≤10초) · Cheerful · Comfortable · Clear
- Streets are public spaces · great for business · can be changed · eco-systems
- Stepped approach: temporary → permanent
- Bioswale · tree pits · rain garden
- Street trees: environmental / social / economic benefits
- Traffic calming: curb extension(모서리), mid-block narrowing(pinch point), chicane, traffic circle, diverter, cul-de-sac
- Parking: parallel 2×6, diagonal 2.5×5